

APSC 131

Week 1 – Knowledge Development Questions Solutions

Lecture 2

Question 1: Write the atomic number, group number and electron configuration for the following elements:

- | | | |
|-------------|-------------|--------------|
| a. Hydrogen | c. Carbon | e. Oxygen |
| b. Lithium | d. Nitrogen | f. Chlorine. |

Hydrogen	H	At.No. 1	$1s^1$
Lithium	Li	At.No. 3	$1s^2 2s^1$
Carbon	C	At.No. 6	$1s^2 2s^2 2p^2$
Nitrogen	N	At.No. 7	$1s^2 2s^2 2p^3$
Oxygen	O	At.No. 8	$1s^2 2s^2 2p^4$
Chlorine	Cl	At.No. 17	$1s^2 2s^2 2p^6 3s^2 3p^5$

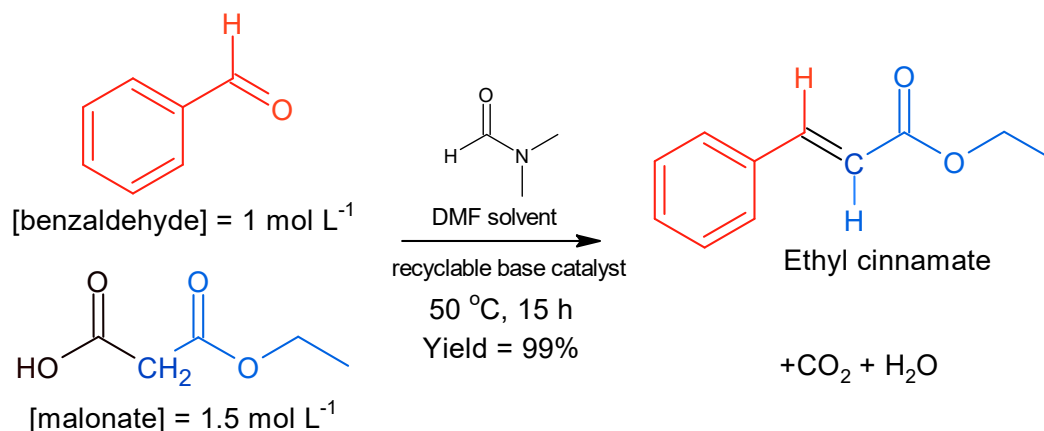
Question 2: Look up the electronegativity of each of the elements in Question 1 and place them in order of most electronegative to least electronegative.

Hydrogen	EN = 2.1
Lithium	EN = 1.0
Carbon	EN = 2.5
Nitrogen	EN = 3.0
Oxygen	EN = 3.5
Chlorine	EN = 3.2

$O > Cl > N > C > H > Li$

Lecture 3

Question 1: Here is an alternate synthesis of ethyl cinnamate that uses a solid, recyclable catalyst.



- a. **Determine** the stoichiometric coefficients of all reactants and products.
 The stoichiometric coefficients for the reactants are both -1, while those of all the products are +1.
- b. **Assess** whether the chemical equation is balanced.
 Since the number of each type of atom is equal in the reactant-side and product-side of the equation, the material balance is satisfied and the equation is balanced.
- c. **Calculate** the composition of the reaction mixture required to produce 1 kg of ethyl cinnamate by this synthesis:
 The yield is 99%. Thus, we require 1% excess limiting reagent. The molar mass of ethyl cinnamate is 176 g/mol.

- i. Mass of aldehyde

The molar mass of benzaldehyde (BA) is 106 g/mol. Thus, the mass of BA needed to produce 1 kg of ethyl cinnamate (EC) is:

$$m_{\text{BA}} = \frac{|v_{\text{EA}}| m_{\text{EC}}}{|v_{\text{BA}}| M_{\text{EC}} \% \text{ yield}} \cdot \frac{100 \%}{M_{\text{BA}}} = \frac{|-1|}{|1|} \cdot \frac{1000 \text{ g}}{176 \text{ g/mol}} \cdot \frac{100 \%}{99 \%} \cdot 106 \text{ g/mol} = 608 \text{ g}$$

- ii. Mass of malonate

The molar mass of malonate (M) is 132 g/mol. Thus, the mass of M needed to produce 1 kg of ethyl cinnamate (EC) is:

$$m_{\text{M}} = \frac{|v_{\text{M}}| m_{\text{EC}}}{|v_{\text{EC}}| M_{\text{EC}} \% \text{ yield}} \cdot \frac{100 \%}{M_{\text{M}}} = \frac{|-1|}{|1|} \cdot \frac{1000 \text{ g}}{176 \text{ g/mol}} \cdot \frac{100 \%}{99 \%} \cdot 132 \text{ g/mol} \cdot \frac{1.5}{1} = 1140 \text{ g}$$

- iii. Approximate volume of DMF solvent

To create a 1 mol/L solution of BA with 608 g of BA, the necessary volume of DMF is:

$$V_{\text{DMF}} = \frac{n_{\text{BA}}}{c_{\text{BA}}} = \frac{\frac{m_{\text{BA}}}{M_{\text{BA}}}}{c_{\text{BA}}} = \frac{\frac{608 \text{ g}}{106 \text{ g/mol}}}{1 \text{ mol/L}} = 5.74 \text{ L}$$

- d. **Estimate** the atom economy of this synthesis and compare its efficiency with that of the lecture example.

$$\text{Atom Economy} = \frac{\text{theoretical product mass}}{\text{corresponding mass of all reagents}} = \frac{m_{\text{EC}} \frac{100 \%}{\% \text{ yield}}}{m_{\text{IB}} + m_{\text{EA}} + m_{\text{PC}}} = \frac{1000 \text{ g} \frac{100 \%}{99 \%}}{608 \text{ g} + 758 \text{ g}} = 0.74$$